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## **Effect of Anisotropic Rock Compaction on Unconventional Well Performance**

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### **Abstract**

Unconventional hydrocarbon reservoirs, particularly shale, exhibit stress-dependent flow properties like permeability and porosity, which are traditionally measured by applying confining stresses on core plugs. This study aims to investigate the impact of anisotropic stress conditions on the flow properties (permeability and porosity) of unconventional reservoirs, specifically shale, and their subsequent effect on well performance and recovery estimation. An advanced laboratory procedure was employed, applying tri-axial anisotropic stresses to shale samples to measure stress-dependent permeability and porosity.

Dynamic simulations, based on these lab measurements, were used to assess the effect of varying stress conditions on well performance. The simulations focused on rock composition, particularly organic matter content, and its influence on permeability decline and rock compressibility. The workflow aimed to predict well performance under varying stress regimes including vertical and horizontal anisotropic stresses. The laboratory results confirmed significant stress dependence in shale permeability and porosity, with pore size reduction observed under high net stress. Numerical simulations, incorporating these stress-dependent properties, indicated a substantial impact on reservoir performance. Variations in flow properties, driven by stress anisotropy, led to considerable differences in well performance predictions.

Simulation results highlighted that ignoring the stress sensitivity of shale's flow properties could lead to inaccurate performance forecasts. By integrating lab data with numerical models, the study showed that incorporating anisotropic stress conditions improves the accuracy of production forecasting, offering more reliable insights into well recovery and performance.

This study introduces a comprehensive integration of advanced lab measurements with numerical simulations to model the impact of anisotropic stress-dependent permeability and porosity on unconventional reservoir performance, providing new insights into more accurate well performance and recovery predictions.

### **Introduction**

In the development of unconventional resource plays, one of the most critical factors influencing production forecasting and field development planning is the ability to accurately model the geomechanical behavior

of the reservoir. Unlike conventional reservoirs where permeability tends to remain relatively stable, production in shale reservoirs is strongly influenced by dynamic permeability reduction caused by in-situ stress disturbance during depletion. This is particularly true in formations with laminated and fine-grained characteristics, such as organic-rich shales, where the stress-dependent nature of both the matrix and fracture systems significantly affects fluid flow. Importantly, these stress conditions are rarely isotropic in real field settings, leading to notable differences in flow behavior when anisotropy is considered.

A pivotal study by [Liu et al. \(2024\)](#), emphasized the role of anisotropic stress sensitivity in determining early-time production from shale wells. Using an Advanced Rock Mechanics System (ARMS), a set of laboratory experiments were conducted to examine how permeability responds to variations in deviatoric stress under different orientations. Their work led to the introduction of a stress sensitivity parameter,  $\gamma$ , which quantifies the effect of deviatoric stress on effective stress. The laboratory findings demonstrated that stress normal to bedding planes—often associated with vertical stress—exerts a much stronger influence on permeability degradation compared to stress aligned with the bedding. Their generalized analytical model further showed that inclusion of anisotropic stress effects could enhance cumulative production estimates, thereby underscoring the importance of including realistic stress configurations in predictive models.

However, despite the insights provided by analytical solutions, these models inherently simplify the problem by assuming conditions such as constant bottom-hole pressure, homogeneous rock properties, and linear flow mechanisms. They typically lack the ability to capture complex phenomena such as multiphase flow behavior, capillary pressure effects, or spatial heterogeneity in rock properties and stress distribution. To address these limitations, this study builds upon Liu et al.'s analytical framework and transitions toward a full-physics, dynamic simulation approach. A single-well numerical model was developed that preserves the rock properties and stress sensitivity values observed in the laboratory while allowing for dynamic, transient modeling of pressure depletion, fluid flow, and mechanical changes throughout the reservoir.

The primary goal of this research is to evaluate the effect of varying degrees of anisotropic stress—specifically the increase of vertical stress relative to lateral stresses—on production behavior and estimated ultimate recovery (EUR). By using a detailed numerical model, this study bridges the gap between lab-scale measurements and field-scale predictions, offering a scalable methodology to incorporate laboratory-derived geomechanical parameters into reservoir simulation models. This work not only validates the laboratory observations at a field-representative scale but also provides a practical workflow for integrating anisotropic stress effects into reservoir simulation frameworks, ultimately aiding in the design of more accurate and robust development plans for unconventional resources.

## Background and Literature Review

### Stress-Dependent Permeability in Shale Reservoirs

The effect of stress on rock permeability in unconventional reservoirs has been widely studied and is well recognized as a key factor influencing production performance. Organic-rich shales, in particular, exhibit a strong coupling between mechanical deformation and fluid flow due to their complex laminated microstructure and high clay or kerogen content. Numerous experimental investigations using triaxial core testing have confirmed that permeability in shales typically declines exponentially with increasing effective stress. This reduction becomes especially pronounced when the applied stress is oriented perpendicular to the bedding planes, further highlighting the anisotropic nature of shale permeability ([Liu et al., 2024](#)).

[Liu et al. \(2024\)](#) proposed an analytical expression that characterizes the permeability  $k$  of shale under varying effective stress conditions:

$$k = k_0 \exp(-\beta \sigma_e) \quad (1)$$

Here,  $k_0$  is the initial permeability at zero effective stress,  $\beta$  is a material-specific stress sensitivity coefficient, and  $\sigma_e$  is the generalized effective stress. This exponential formulation is consistent with several

earlier empirical models (e.g., [Bhandari et al., 2015](#)) and has been widely applied in both laboratory and field modeling workflows.

A similar exponential relationship is also reported for porosity:

$$\phi = \phi_0 \exp(-\alpha \sigma_c) \quad (2)$$

In this equation,  $\phi_0$  is the initial porosity at zero stress, and  $\alpha$  is the porosity stress-sensitivity coefficient. Notably, in typical shale formations, the magnitude of  $\alpha$  is approximately one order of magnitude smaller than  $\beta$ , suggesting that permeability is significantly more sensitive to stress changes than porosity. This observation supports the focus of this study on stress-dependent permeability as the dominant control on production degradation in unconventional reservoirs, [Abdelrahman et al. \(2024\)](#).

To account for anisotropic stress effects, [Liu et al. \(2024\)](#) redefined the effective stress as follows:

$$\sigma_e = \sigma_c + \gamma \sigma_d - \alpha p \quad (3)$$

Where:

- $\sigma_c$ : Confining (vertical) stress
- $\sigma_d$ : Deviatoric stress
- $\gamma$ : Deviatoric-stress sensitivity parameter
- $\alpha$ : Biot's coefficient
- $p$ : Pore pressure

The deviatoric stress term  $\sigma_d$  is defined as the difference between axial and confining stress:

$$\sigma_d = \sigma_a - \sigma_c \quad (4)$$

Where  $\sigma_a$  represents the axial stress, often associated with horizontal stresses in the subsurface (i.e., minimum and maximum horizontal stress components). [Figure 1](#) illustrates this stress configuration in a cylindrical rock core under laboratory conditions.

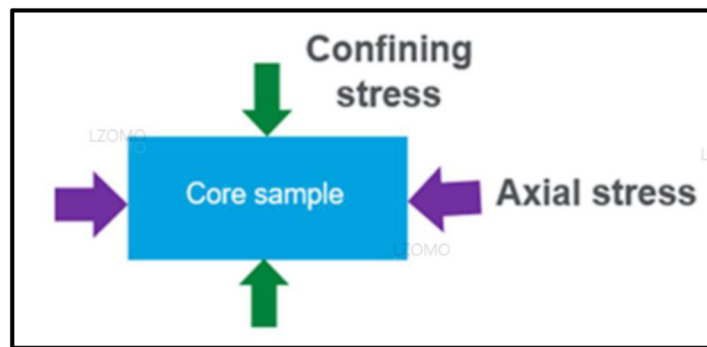


Figure 1—Schematics of stresses for a cylindrical rock sample under a confining stress and an axial stress in a laboratory setting. (Adapted from [Liu et al., 2024](#))

During the experimental phase, Liu et al. employed the Advanced Rock Mechanics System (ARMS), a unique laboratory setup that combines a high-capacity loading frame with a nano-permeameter. This system enabled precise application of anisotropic stress conditions while continuously monitoring permeability changes across varying orientations. The test data provided a robust basis for determining the  $\gamma$  parameter by fitting measured values to [equations \(1\) through \(4\)](#). The results revealed that  $\gamma$  varied within a specific range across different rock types, and these values were used as inputs for the dynamic simulations developed in this study to assess the impact of low, average, and high anisotropic stress sensitivity on well performance.

## Analytical Models for Anisotropic Effects

Liu et al. (2024) applied advanced laboratory measurements to a theoretical framework, developing an analytical model to assess how anisotropic stress conditions affect early cumulative production from shale reservoirs. This model incorporated stress-sensitive permeability relationships derived from laboratory experiments conducted under anisotropic conditions, where stress perpendicular to the bedding plane had a disproportionately large effect on permeability.

A key result of their analytical work was the derivation of a dimensionless cumulative production expression, defined as:

$$Q^* = \exp\left\{-\beta\left((1-2\gamma)\sigma_v + \gamma(\sigma_{H1} + \sigma_{H2})\right)\right\}(T)^{1/2} \quad (5)$$

Where:

- $Q^*$ : Dimensionless cumulative hydrocarbon production
- $\beta$ : Stress sensitivity coefficient of permeability
- $\gamma$ : Deviatoric-stress sensitivity parameter
- $\sigma_v$ : Vertical (confining) stress
- $\sigma_{H1}, \sigma_{H2}$ : Maximum and minimum horizontal stresses
- $T$ : Dimensionless time (normalized by a reference time  $T_{ref}$ )

This model accounts for the directional nature of stress in shale formations, capturing how higher  $\gamma$  values (i.e., increased stress anisotropy) negatively affect hydrocarbon flow by amplifying permeability degradation.

Importantly, the term  $(1-2\gamma)\sigma_v + \gamma(\sigma_{H1} + \sigma_{H2})$  generalizes the effective stress concept to consider deviatoric effects, with  $\gamma$  controlling how much weight is given to horizontal vs. vertical stress.

When  $\gamma=0$ , the system is isotropic, and effective stress is dominated by vertical stress. When  $\gamma>0$ , anisotropy is introduced, with more influence from horizontal stresses. This simple, elegant model provided a powerful theoretical insight: even small deviations from isotropic stress assumptions can result in measurable changes in production forecasts.

Liu's analytical approach, however, carries limitations that are inherent to most closed-form solutions. It assumes:

- Single-phase flow (ignores condensate dropout or liquid saturation effects)
- Homogeneous rock and fluid properties
- Constant bottom-hole pressure

These simplifications limit the model's ability to capture the dynamic, nonlinear, and multiphase nature of production in real reservoirs. In particular, the interaction between permeability degradation and condensate banking, and the evolving stress field near hydraulic fractures, are not represented. These limitations motivated the current study to transition from analytical to numerical simulation using a more rigorous full-physics model.

## Dynamic Simulation with Stress-Dependent Parameters

To overcome the constraints of the analytical approach, Abdelrahman et al. (2024) previously developed a full-physics dynamic simulation workflow. This workflow integrates lab-measured stress sensitivity parameters directly into numerical reservoir models and tracks how permeability and porosity evolve with effective stress during depletion. Their study, which focused on unconventional multi-well developments,

found that stress-dependent permeability was a dominant control on production decline and had a more significant effect on EUR than porosity degradation.

Inspired by that framework, the current study focuses on a simplified but detailed single-well model using gas condensate properties. It specifically aims to isolate the effects of anisotropic stress—modeled through varying  $\gamma$  values—on permeability evolution, pressure depletion, and ultimate recovery. The workflow includes the same exponential decay formulations for permeability and porosity:

$$k = k_0 \exp(-\beta \sigma_e) \quad (1)$$

$$\phi = \phi_0 \exp(-\alpha \sigma_e) \quad (2)$$

Where:

$$\sigma_e = \sigma_c + \gamma \sigma_d - \alpha p \quad (\text{Eq. 3 from Liu et al., 2024})$$

$$\sigma_d = \sigma_a - \sigma_c \quad (\text{Eq. 4})$$

The numerical model allows us to simulate how these relationships play out dynamically, as pressure depletes and effective stress increases. It also enables us to track multiphase flow, phase changes, and changes in local permeability around the fracture over time.

This approach ensures that the analytical insights from Liu et al. (2024) are honored while expanding the model's realism, giving us more actionable insights for field-scale development.

## Methodology

### Dynamic Model Configuration

A conceptual single-well model was constructed to evaluate the effect of anisotropic stress regimes on well performance. The model uses a Cartesian grid structure with local grid refinement (LGR) near the hydraulic fracture to better capture flow behavior from the stimulated rock volume (SRV). As shown in Figure 2, the trilinear modeling approach developed by Brown et al. (2009) and Ozkan et al. (2011) was adopted to divide the reservoir into three flow regions:

- **Main hydraulic fractures (EPV):** Effective Propped Volume through which most of the fluid initially flows
- **SRV:** Stimulated rock volume adjacent to the fracture
- **Matrix:** Unstimulated, low-permeability formation

This discretization enables the simulation to capture the varying pressure differentials and compaction effects across these regions, which are especially important under different stress regimes.

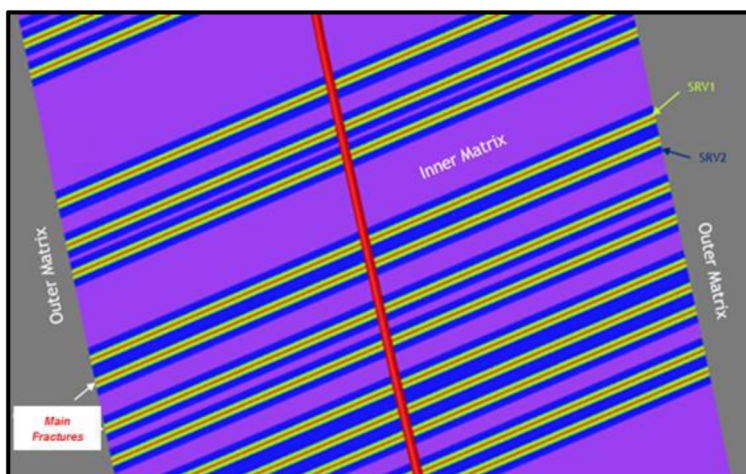


Figure 2—Map view of the LGR regions using the trilinear modeling approach

To configure the hydraulic fracture geometry in the static model, a commercial fracture simulation software was used to model a representative stimulation design. This includes fracture half-length, height, and conductivity. As shown in Figure 3, these fracture attributes were then explicitly embedded into the grid using a novel LGR mapping workflow that ensures accurate spatial placement and conductivity representation. For simplicity and computational efficiency, only one hydraulic fracture stage was modeled, and production results were later scaled to represent the contribution from multiple stages in a full horizontal well.

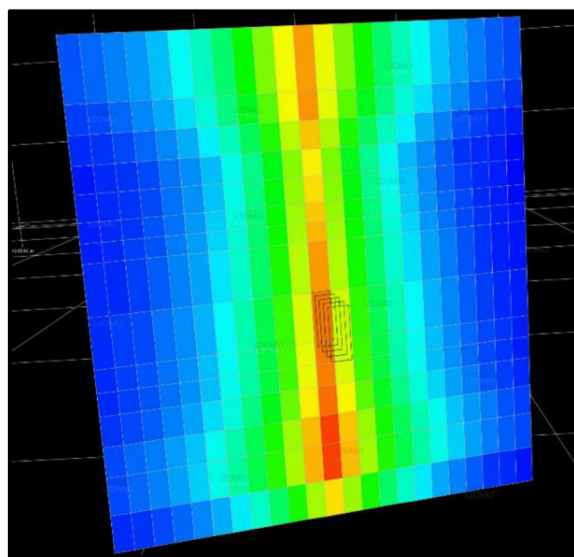


Figure 3—Fracture model results mapped into the static model

### Fluid Characterization (EOS)

To accurately represent fluid flow behavior in the reservoir, a gas condensate system was modeled using a tuned Equation of State (EOS). The fluid composition was initialized to match field conditions, incorporating reservoir temperature, pressure, and gas-to-condensate ratio (GOR). These values were derived from a representative producing well in the asset.

The tuned EOS was then used to populate the compositional model input, enabling proper representation of phase changes—particularly condensate dropout—as pressure declines below the dew point. This step is

essential for capturing multiphase flow behavior in gas condensate reservoirs and adds realism compared to dry gas or single-phase models often used in analytical work.

## Rock Physics and Geomechanical Parameters

**Relative Permeability and Capillary Pressure.** Relative permeability curves were estimated using the Corey correlation, based on typical values for tight gas-condensate systems. Given the very low permeability nature of the rock, the model incorporated a steep gas relative permeability decline at higher liquid saturations, consistent with field observations of condensate banking effects. Capillary pressure was included but kept constant across cases to isolate the impact of stress-dependent permeability.

**Stress-Dependent Parameters.** The following geomechanical inputs were considered:

- **Permeability stress sensitivity coefficient ( $\beta$ ):** Determined from ARMS core tests (Liu et al., 2024)
- **Porosity stress sensitivity coefficient ( $\alpha$ ):** An order of magnitude lower than  $\beta$
- **Biot's coefficient ( $\alpha$ ):** Liu et al. (2024) demonstrated that the Biot's coefficient  $\alpha$  is close to one for the studied samples; in this study we use also  $\alpha = 1$
- **Confining and axial stresses ( $\sigma_c$ ,  $\sigma_a$ ):** Representing vertical and horizontal in-situ stresses
- **Deviatoric-stress sensitivity parameter ( $\gamma$ ):** A critical variable that quantifies anisotropic behavior

Each of these parameters feeds into the effective stress calculation, influencing how permeability evolves with pore pressure depletion.

To visualize the effect of different  $\gamma$  values on permeability, compaction isotherms were generated for each case. These isotherms depict permeability degradation as a function of effective stress for the base case ( $\gamma = 0$ ) and for low, medium, and high  $\gamma$  scenarios. The porosity degradation was held constant across all cases to avoid bias and focus solely on the impact of directional stress effects on permeability. Figure 4 illustrates the resulting compaction isotherms, highlighting the accelerating loss of permeability with increasing  $\gamma$ .

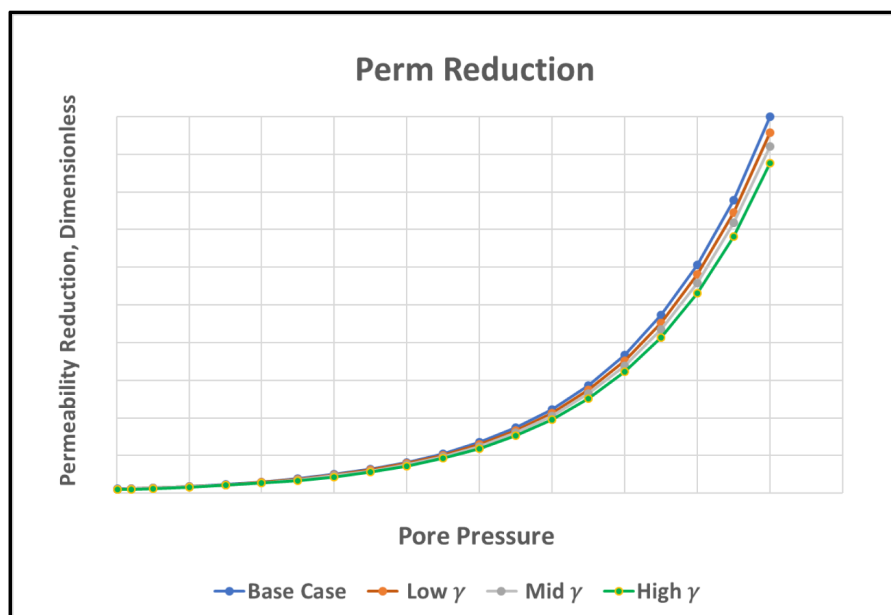


Figure 4—Rock Compaction Isotherms—Permeability vs. Pore Pressure Change for various  $\gamma$  values

## Base Case Definitions

The simulation base case was designed as a single horizontal well intersected by one hydraulic fracture stage. The objective of the base case was to establish a reference point for evaluating how anisotropic stress (captured through  $\gamma$ ) influences production behavior and EUR.

For the base case, an isotropic stress regime was assumed, corresponding to  $\gamma = 0$ . The permeability and porosity degradation were modeled using the exponential relations in Equations (1) and (2), incorporating the  $\alpha$  and  $\beta$  values calibrated from lab measurements. This isotropic scenario allowed the well to produce under a specified target gas rate while maintaining a minimum allowable bottom-hole pressure. The base case simulation was allowed to run until production declined to economic limits, and the well's Estimated Ultimate Recovery (EUR) was calculated.

Figure 5 presents the base case simulation results, including oil production rate (top left), cumulative oil production rate (top right), gas production rates (bottom left), and water production rate (bottom right).

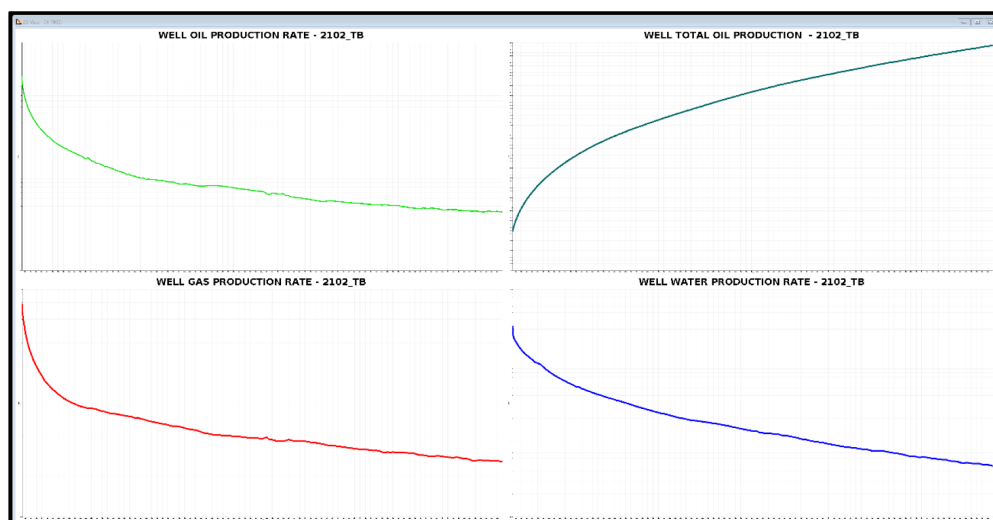


Figure 5—Base Case Simulation Results: Oil, Water, & Gas Production Rate, and Cumulative Oil Recovery For Base Case ( $\gamma = 0$ )

Following the base case, three additional simulations were conducted, each representing a different level of anisotropy by varying  $\gamma$  (low, medium, and high values). These  $\gamma$  values were selected based on the range reported in Liu et al. (2024) and corresponded to field-measured variations in stress response. For all these cases, the same porosity compaction curve was used as in the base case, thereby isolating the impact of permeability analysis.

## Results and Discussion

### Pressure and Permeability Evolution

The simulation results confirm that permeability near the hydraulic fracture declines rapidly as production begins, due to the increasing effective stress caused by pressure depletion. As the pore pressure drops, the differential between in-situ stresses and pore pressure increases, leading to compaction and a corresponding loss of permeability. This effect becomes more pronounced when deviatoric stress is introduced through a non-zero  $\gamma$  value.

Figure 6 presents a map view of the pressure distribution surrounding the fracture at a specific timestep, highlighting the non-uniform depletion patterns that evolve under different stress anisotropy conditions. In the anisotropic cases, pressure depletion is more localized, and the permeability reduction extends further into the SRV region, and accelerating production decline.

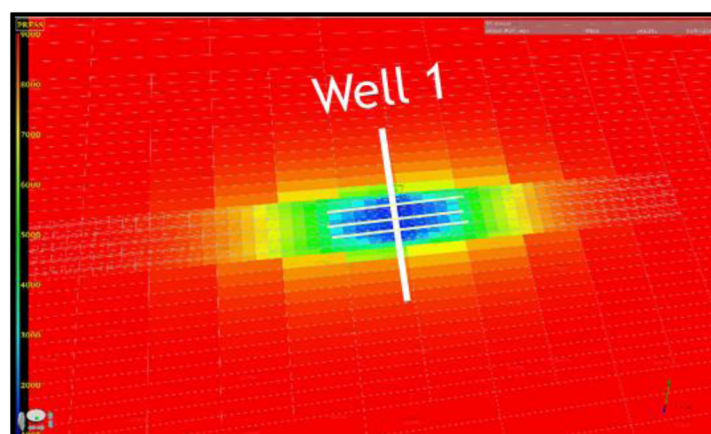


Figure 6—Map view of pressure depletion at one hydraulic fracture stage under anisotropic stress

In scenarios with elevated  $\gamma$  values, permeability near the fracture faces degraded faster than in the base case, resulting in restricted flow paths and reduced connectivity to the matrix. These results illustrate how directional stress sensitivity not only influences the magnitude of permeability but also its spatial distribution, creating zones of higher resistance that hinder reservoir drainage efficiency over time.

### EUR Sensitivity

The estimated ultimate recovery (EUR) of the well was computed for all cases to quantify the impact of anisotropic stress on long-term performance. The base case ( $\gamma = 0$ ), which assumes isotropic stress, yielded the highest EUR. As  $\gamma$  increased, representing more pronounced stress anisotropy, the EUR declined progressively.

Figure 7 shows the EUR comparison between the base case and the three anisotropic cases, corresponding to low, medium, and high  $\gamma$  values. The results reveal a clear and consistent trend: as  $\gamma$  increases, the well's recovery factor decreases. This confirms the analytical predictions made by Liu et al. (2024), while extending the analysis to field-scale, full-physics conditions.

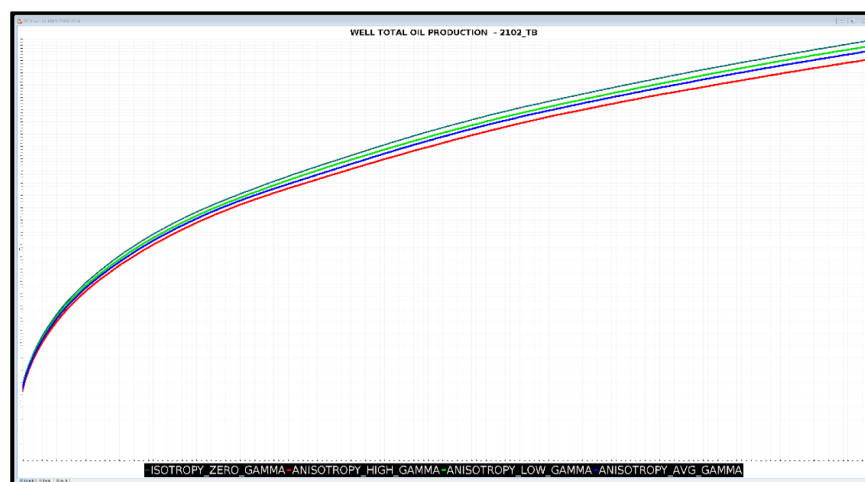


Figure 7—Comparison of well EUR for base case vs. low, mid, and high  $\gamma$  scenarios

The results also support findings by Abdelrahman et al. (2024) that stress-dependent permeability, more so than porosity, is the dominant driver of production degradation in unconventional formations. The observed variation in EUR solely due to permeability confirms the critical importance of geomechanical integration in unconventional reservoir simulation.

## Implications for Field Development

### Completion and Fracture Design

The findings from this study underscore the need to consider in-situ stress anisotropy during hydraulic fracture design and completion optimization. In formations with high vertical stress or pronounced stress gradients, vertical fracture growth may be limited, and near-wellbore conductivity may degrade more rapidly due to compaction effects. Completion strategies that do not account for these directional stress sensitivities risk suboptimal fracture geometry and inefficient stimulation.

This suggests that fracture design should be adapted to local stress regimes, with consideration given to proppant selection, pump schedules, and stage spacing to mitigate the effects of anisotropic compaction.

### EUR Forecasting

Conventional EUR forecasting methods, especially those that rely on type curves or analytical solutions assuming isotropic stress, may overestimate recovery in formations with significant stress anisotropy. The results of this study provide evidence that incorporating deviatoric stress sensitivity into simulation workflows can lead to more accurate recovery predictions. This has direct implications for reserves booking, economic modeling, and development planning.

### Integration of Lab and Field Data

The workflow presented here offers a pathway for integrating laboratory-derived geomechanical parameters into field-scale models. By using parameters like  $\beta$  and  $\gamma$ , which can be measured on core plugs, reservoir engineers can bridge the gap between small-scale rock testing and full-field simulation. When combined with regional stress models or in-situ stress logs, this approach can improve geomechanical characterization across the basin and enable more informed development decisions.

## Conclusions

This study presents a numerical investigation into the impact of anisotropic stress sensitivity on well productivity in a gas condensate unconventional reservoir. Building on prior analytical work by [Liu et al. \(2024\)](#), we applied laboratory-derived geomechanical parameters in a full-physics, single-well dynamic simulation model to evaluate permeability degradation under different stress anisotropy conditions.

The key conclusions are:

1. Anisotropic stress conditions significantly impact reservoir performance.  
Higher deviatoric-stress sensitivity ( $\gamma$ ) leads to accelerated permeability degradation near the fracture, resulting in reduced EUR.
2. Numerical models extend analytical insights by incorporating multiphase flow and time-dependent behavior.  
The transition from a simplified analytical framework to a compositional, stress-sensitive numerical model allows for more realistic prediction of well performance.
3. Stress-dependent permeability is a dominant control on production decline.  
Holding porosity constant across scenarios allowed this study to isolate the effect of permeability, confirming that it exerts greater influence on EUR in stress-sensitive formations.
4. Field development plans must integrate anisotropic geomechanics.  
Completion design, stimulation treatment, and recovery forecasting should account for the directionality of stress effects to avoid over- or underestimation of well potential.
5. Lab-to-field workflows are viable and beneficial.  
Using  $\gamma$  and  $\beta$  values from core measurements in reservoir simulation enables a more accurate translation of experimental observations into development strategies.

This study recommends that reservoir simulation workflows—particularly for unconventional gas condensate systems—incorporate anisotropic stress formulations wherever possible, using lab-derived parameters to inform stress-sensitive models. Doing so improves recovery forecasting and supports more robust field development planning.

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